

# Screen Design and Colour



- The screen design is an important part of the UI development
- A poor screen design may degrade user performance
- Screen layout must be carefully designed
- There are numerous guidelines (we have seen already some of them)

# Screen Layout Guidelines

- Several types:

General layout of information

Text 

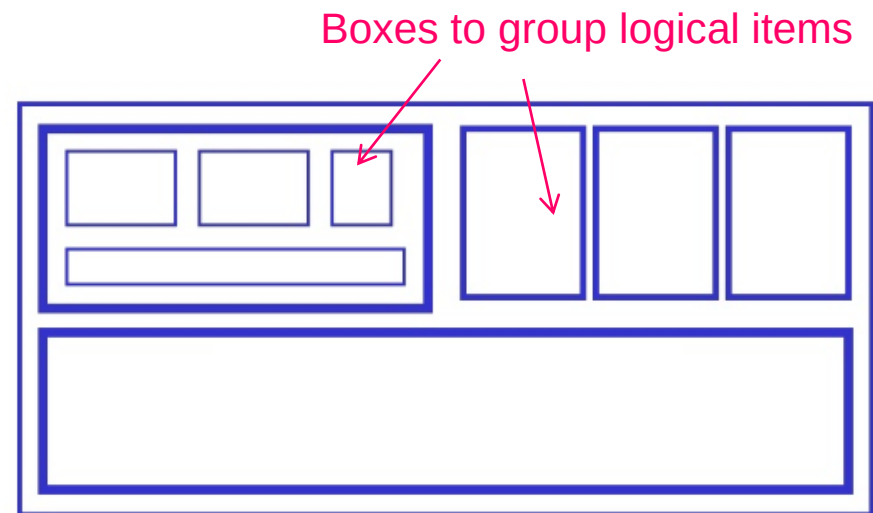
Numbers

Coding techniques (colour and others)

# Information layout

- Include only the needed information
- Include all needed information
- Begin at the top left corner and align left (in Western culture)
- Group items according to type
- Leave plenty of white space
- Use leaders in multiple columns

**This is related to how humans  
analyse an image**



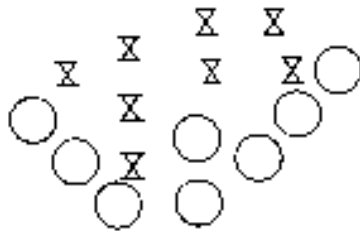
# Gestalt “Laws”

Help understand how visual stimuli in a scene are perceived

XX XX XX

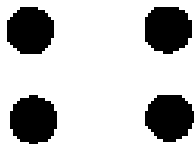
Proximity

Near stimuli are perceived as a group



Similarity

Similar stimuli tend to be grouped  
(may override proximity)



Closure

Stimuli tend to be grouped  
in complete figures



Simplicity

Ambiguous stimuli tend to be resolved  
Using the simplest explanation



Good continuation

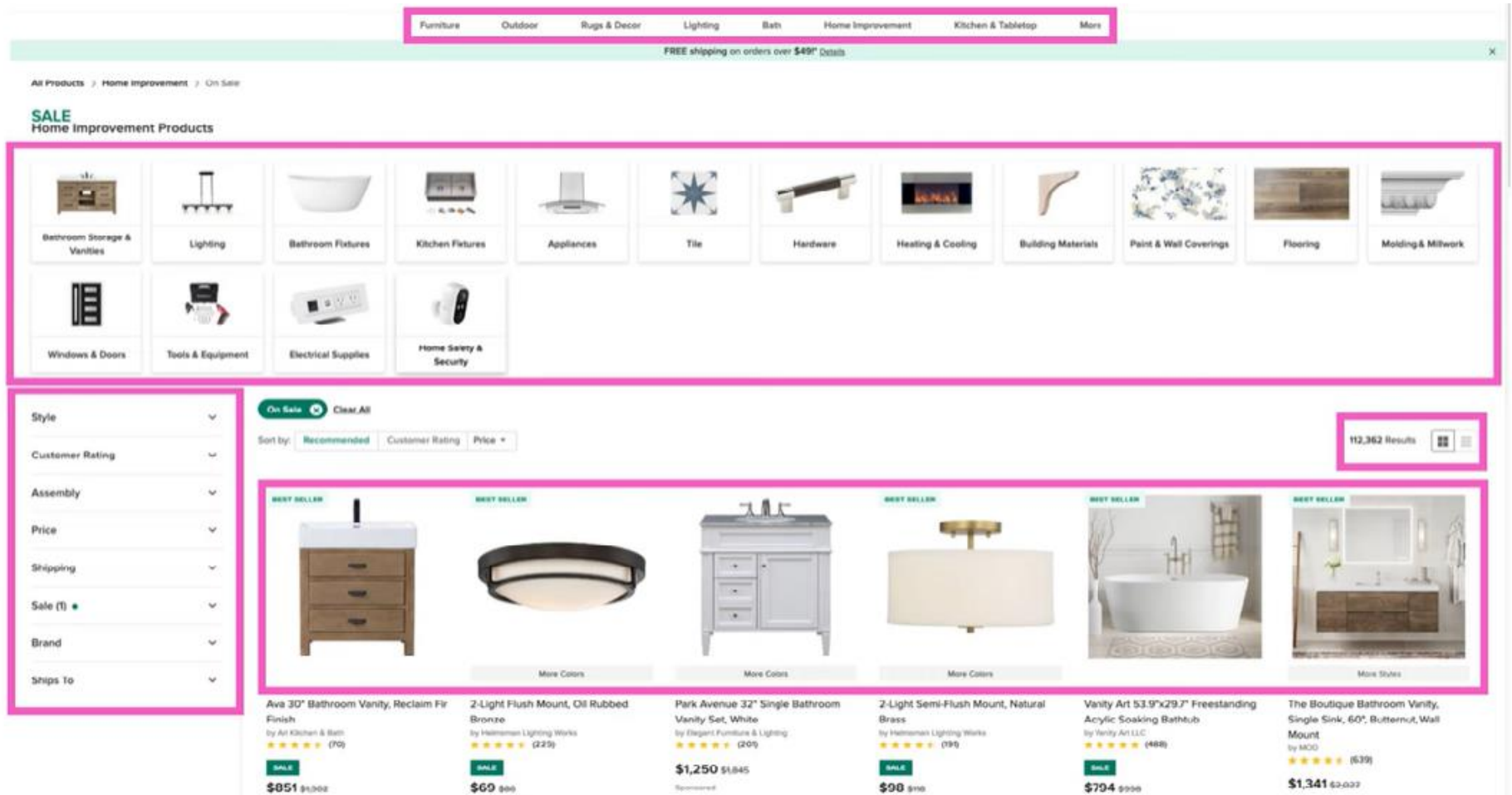
Stimuli tend to be grouped as to minimize  
variations or discontinuities



Symmetry

Regions delimited by symmetric tend  
to be perceived as coherent figures

# Gestalt Laws applied in screen layout design



[The Gestalt Principles for User Interface Design \(Video\)](#)  
[Similarity: Gestalt Principle for User Interface Design](#)

# Some simple rules for screen design

What is similar should be together

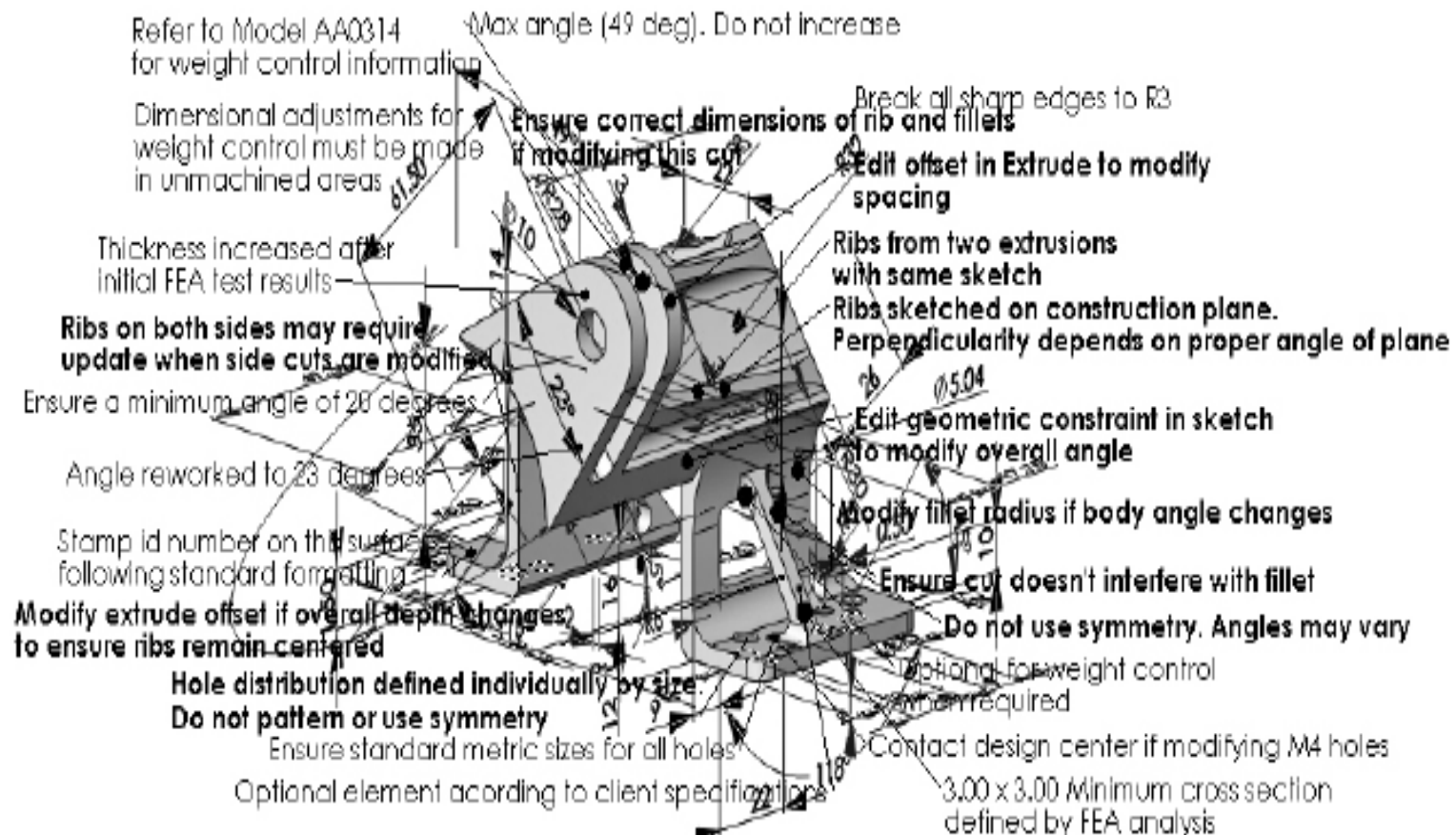
|                                                                   |                                                                   |
|-------------------------------------------------------------------|-------------------------------------------------------------------|
| <b>Billing details:</b><br>Name<br>Address: ...<br>Credit card no | <b>Delivery details:</b><br>Name<br>Address: ...<br>Delivery time |
|-------------------------------------------------------------------|-------------------------------------------------------------------|

---

|                        |          |           |       |
|------------------------|----------|-----------|-------|
| <b>Order details:</b>  |          |           |       |
| item                   | quantity | cost/item | cost  |
| size 10 screws (boxes) | 7        | 3.71      | 25.97 |
| .....                  | ...      | ...       | ...   |

# Include only the needed information

## Avoid Visual Clutter



# Text

- Avoid using only capital letters (are more difficult to read)
- Avoid text with many capital letters
- Do not use too many fonts for emphasis
- In multiple columns use leaders or greying

use fonts for emphasis  
(but not too many)

ABCDEF HIJKLM  
NOPQRSTUVWXYZ

|                   |       |       |       |       |       |       |       |       |       |       |
|-------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Alcântara - Terra |       |       | 12:36 |       |       | 13:06 |       |       | 13:36 |       |
| Campolide         | 12:15 |       | 12:41 | 12:45 |       | 13:11 | 13:15 |       | 13:41 | 13:45 |
| Rossio            | 12:19 |       |       | 12:49 |       |       | 13:19 |       |       | 13:49 |
| Sete Rios         |       | 12:19 | 12:43 |       | 12:49 | 13:13 |       | 13:19 | 13:43 |       |
| Entrecampos       |       | 12:22 | 12:47 |       | 12:52 | 13:17 |       | 13:22 | 13:47 |       |
| Roma - Areeiro    |       | 12:24 | 12:49 |       | 12:54 | 13:19 |       | 13:24 | 13:49 |       |

Use greying

Willy Wonka and the Chocolate Factory  
Winston Churchill - A Biography  
Wizard of Oz  
Xena - Warrior Princess



Left aligned: more readable

Willy Wonka and the Chocolate Factory  
Winston Churchill - A Biography  
Wizard of Oz  
Xena - Warrior Princess



Right aligned: fine for effects  
But more difficult to read

In multiple columns it is difficult to read across gaps:

|                |     |
|----------------|-----|
| sherbert       | 75  |
| toffee         | 120 |
| chocolate      | 35  |
| fruit gums     | 27  |
| coconut dreams | 85  |

use leaders

|                |       |     |
|----------------|-------|-----|
| sherbert       | _____ | 75  |
| toffee         | _____ | 120 |
| chocolate      | _____ | 35  |
| fruit gums     | _____ | 27  |
| coconut dreams | _____ | 85  |

or greying

|                |     |
|----------------|-----|
| sherbert       | 75  |
| toffee         | 120 |
| chocolate      | 35  |
| fruit gums     | 27  |
| coconut dreams | 85  |

- Messages shall:
  - Have a detail level adequate to user knowledge and experience
  - Be specific and understandable
  - Be brief and concise
  - Be positive
  - Be helpful

## Error messages

Too verbose

**better**

|                                                              |                                        |
|--------------------------------------------------------------|----------------------------------------|
| The processing of the text editor yielded 23 pages of output | Output 23 pages                        |
| Error in SIZE field                                          | Error: SIZE range is 4 to 16           |
| Cannot exit before saving file                               | Save file before exiting               |
| Bad/illegal file name                                        | Maximum file name length is 8 chars    |
| Syntax error 1542                                            | Unmatched left parenthesis in line 210 |

Too vague

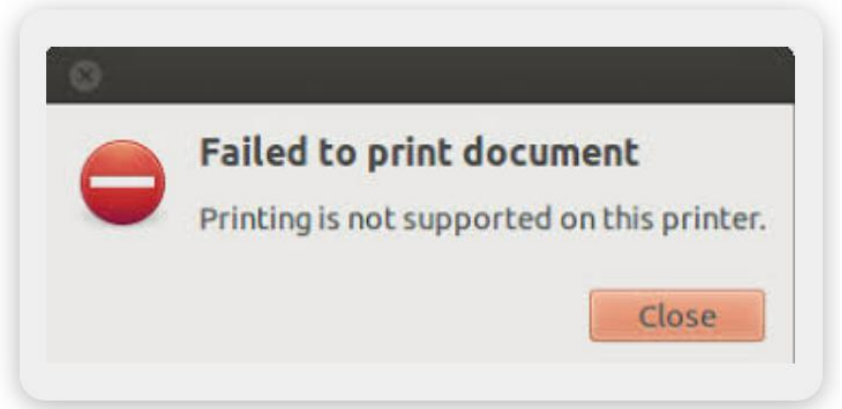
Negative

Not helpful

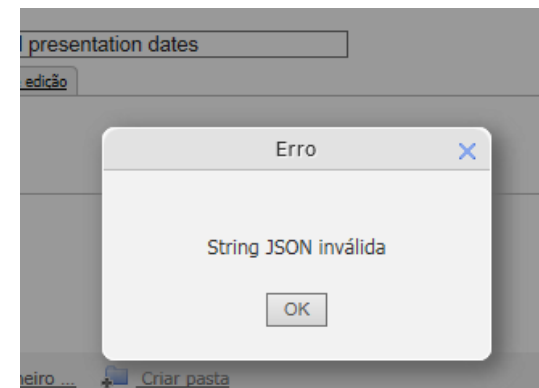
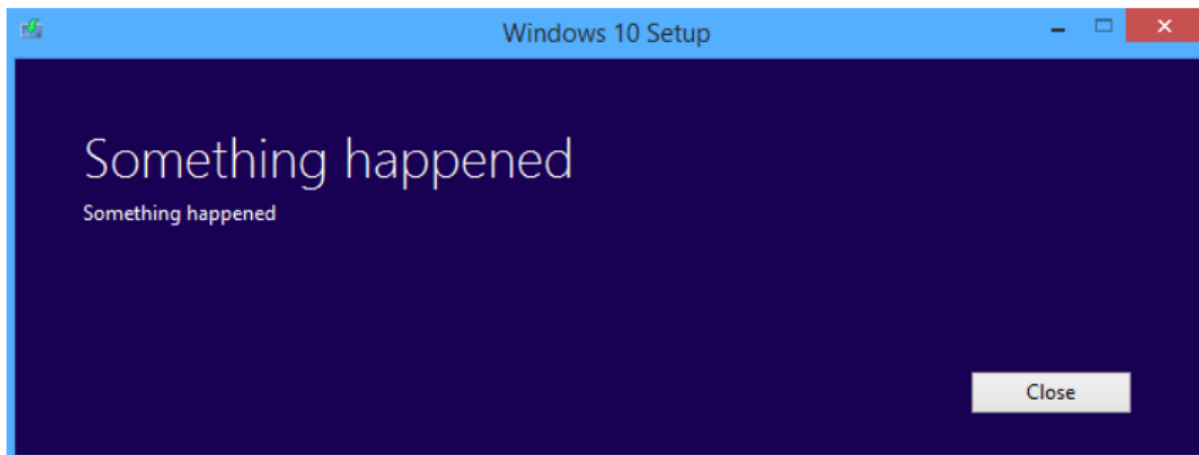
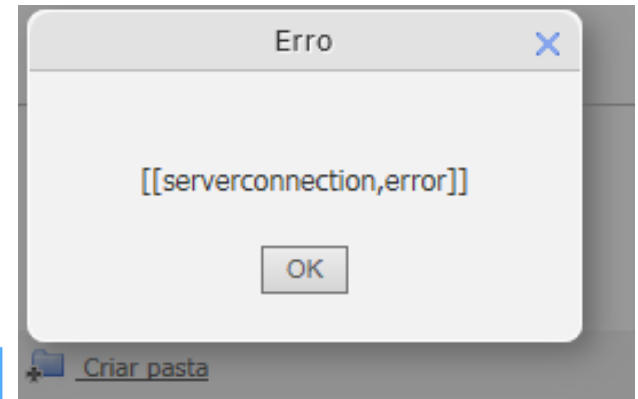
## Examples of useless messages for users



(Except for Chinese speaking people!)



Moodle:



# Numbers

- Integers shall be right justified
- Real numbers shall be aligned by the decimal point
- Avoid unnecessary zeros (at left)
- Long numbers shall be divided in groups



Which is the largest?

532.56  
179.3  
256.317  
15  
73.948  
1035  
3.142  
497.6256

627.865  
1.005763  
382.583  
2502.56  
432.935  
2.0175  
652.87  
56.34

75  
120  
35  
27  
85

Right align integers

Align decimal points

# Numbers

Better

|                                        |                                        |
|----------------------------------------|----------------------------------------|
| 10<br>100<br>1000<br>10000             | 10<br>100<br>1000<br>10000             |
| 100.00<br>25.365<br>5432.01<br>1.45591 | 100.00<br>25.365<br>5432.01<br>1.45591 |
| 10:1 p.m.<br>002                       | 10:02 p.m.<br>2                        |
| 6173954686                             | 617-395-4686                           |

# Coding techniques

Blinking

Bold

Size

Font

Underlining

Shape

Special characters and icons

Proximity

Borders

Sound

Colour

**Main guideline: use parsimoniously any coding technique!**

## 5 Important Factors of Control Room Display Wall Design



What about large screens?



Visual ergonomics: large format screens or multidisplay | GESAB

# Specific problems for large displays

Designing for large screens may be different due:

Size

Viewing angle

Viewing distance

...



# Screen layout for large displays: some guidelines

Take Advantage of the Extra Space:

- Show More Content

- Multi-Column Layouts

- Larger margins and fonts

Prioritize Information Hierarchy is important:

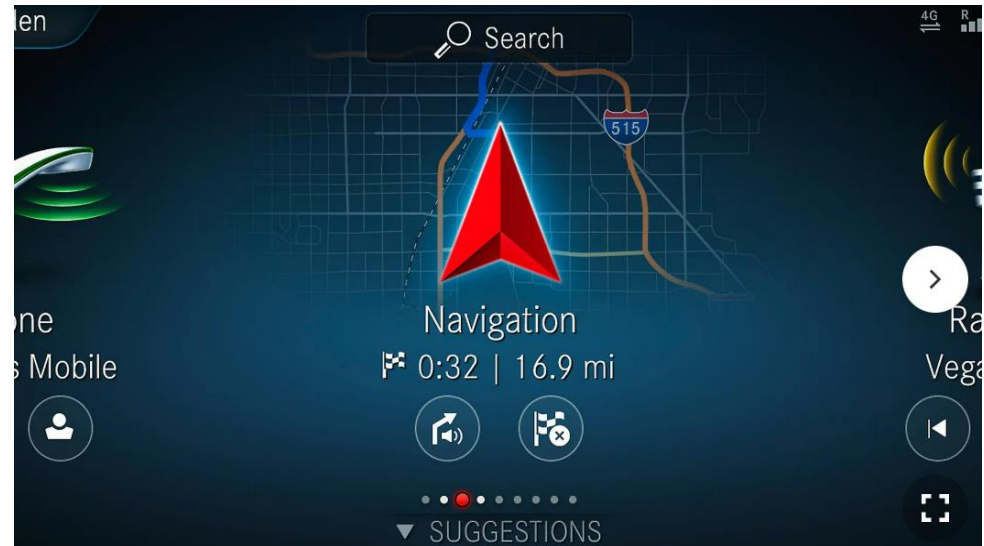
- Visual Composition

- Group Related Elements

- Ensure Reachability

Motion and Animations: slower for larger distances

## Revolution in the Cockpit: Mercedes-Benz UX



What about specific platforms?



## The Role of Wearable Devices in Monitoring Health

# Specific issues for different platforms: mobile

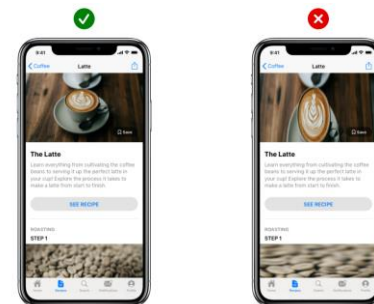
- Many guidelines are similar for mobile and desktop design, but their mobile interpretation is **much more unforgiving**
  - Context of use
  - Size of screen
  - Platform limitations
- There are guidelines for different platforms



[Design & Plan | Android Developers](#)

[UI Design Dos and Don'ts - Apple Developer](#)

[Mobile UX Sharpens Usability Guidelines](#)



# Specific issues for different platforms: wearable

Designing for wearables: some rules:

Glanceability

Keep it simple

Minimalistic interface

...

Remember also:

Privacy and Wireless connectivity

[Designing for wearables: 7 rules and best practices!](#)

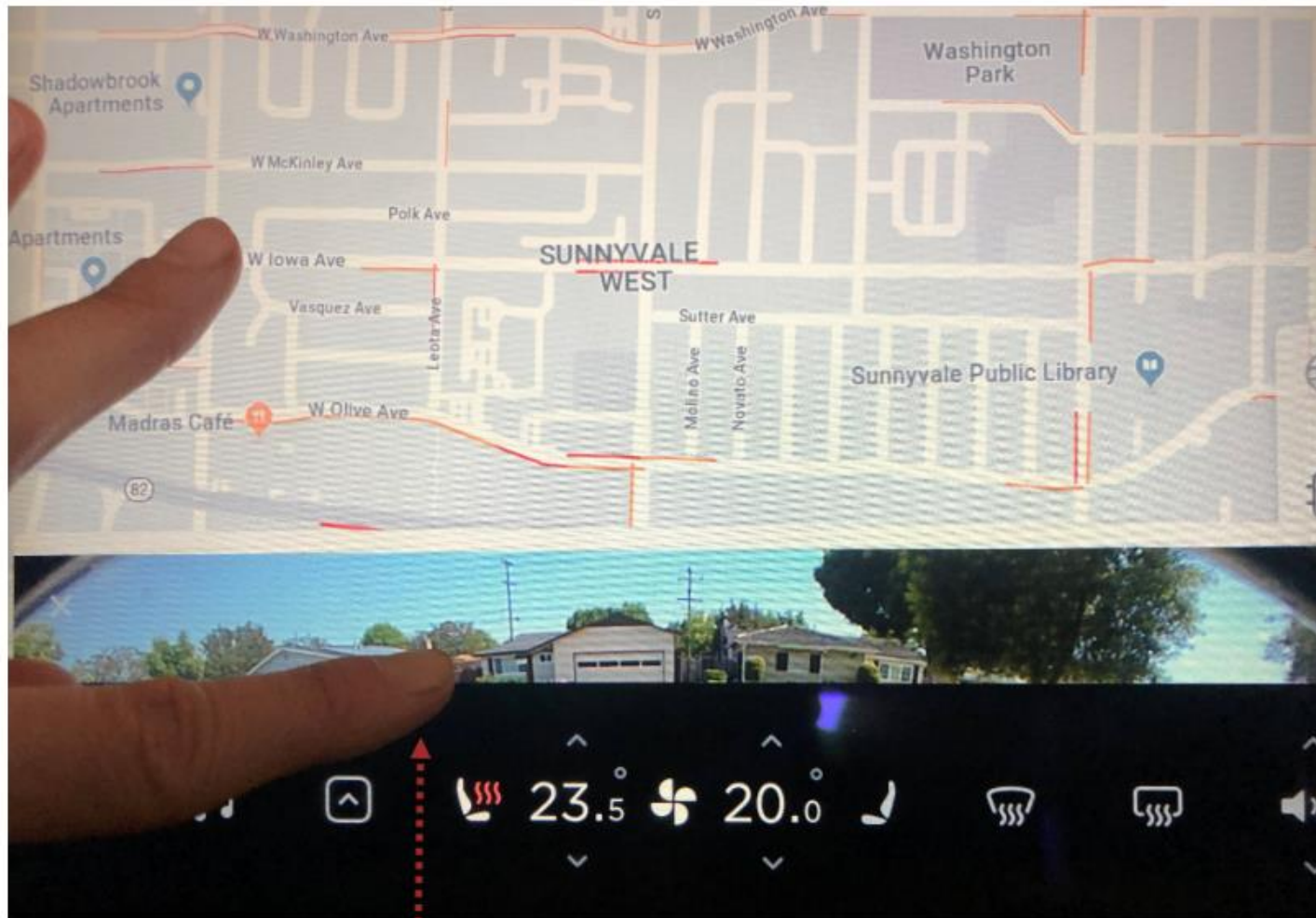


## A critical situation: automotive dashboards



[Automobile Dashboard Technology Is Simply Awful | Scientific American](#)

## The Tesla case:



Tesla's Touchscreen UI: A Case Study of Car-Dashboard User Interface



Primary task:  
Driving  
(Visual attention &  
hand/foot movements)

AND

Secondary task:  
Interface navigation  
(Visual attention &  
hand movements)

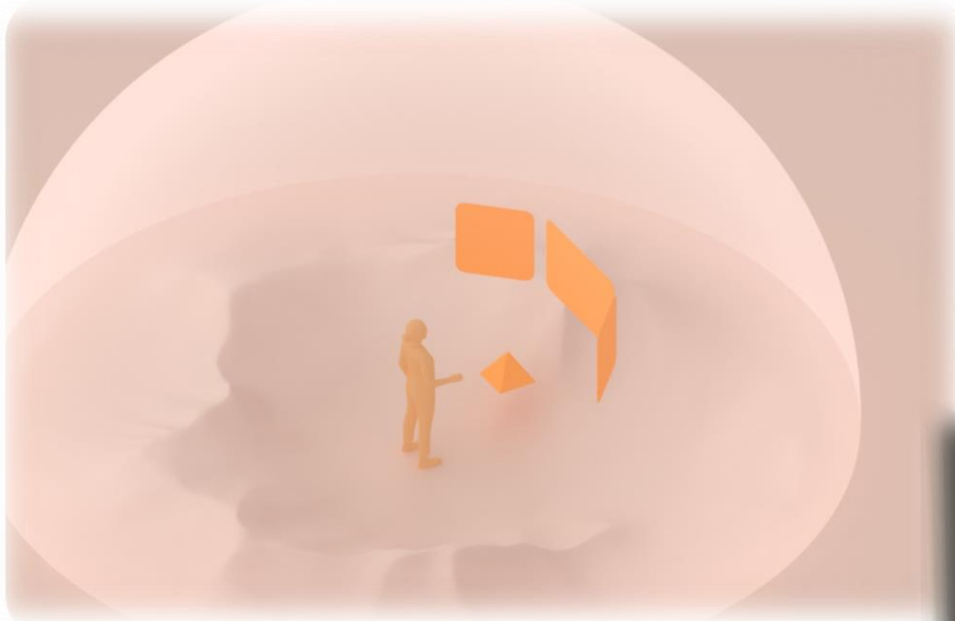


Diverted visual attention  
/ Increased risks of  
accidents

“Manufacturers need to examine whether their in-car display strategies ... are sacrificing other critical automobile functions, particularly safety.”

[Interface-Driven Customer Experience: Redefining User Interface \(UI\) Design for Automotive Infotainment System | IEEE Journals & Magazine | IEEE Xplore](#)

# Specific issues for different platforms: Extended Reality

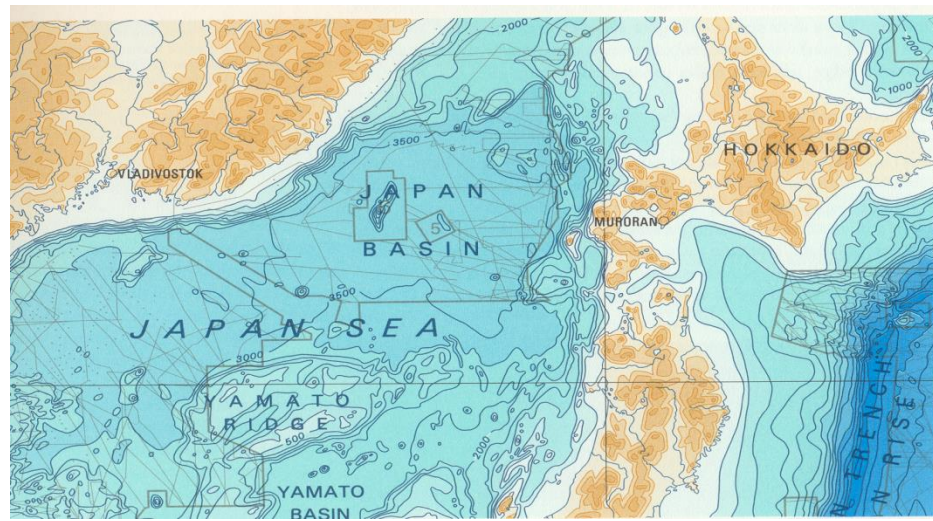


[XR | UI Design | XR Headsets & wired XR Glasses | Android Developers](#)



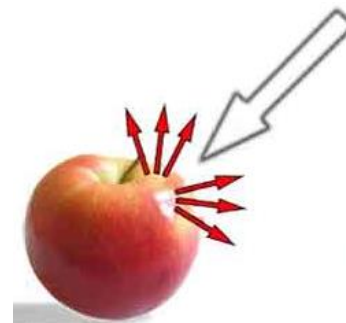
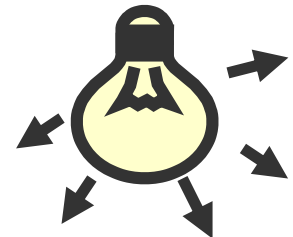
[Immersive experiences | Apple Developer Documentation](#)

# Colour usage



# Colour

- Colour is a **complex and multidisciplinary** subject:
  - Physics
  - Physiology and psychology
  - Art and graphic design
  - Interactive systems design
- The perceived colour of an object depends on the:
  - Material characteristics
  - Illumination
  - Ambient colour
  - Human visual system

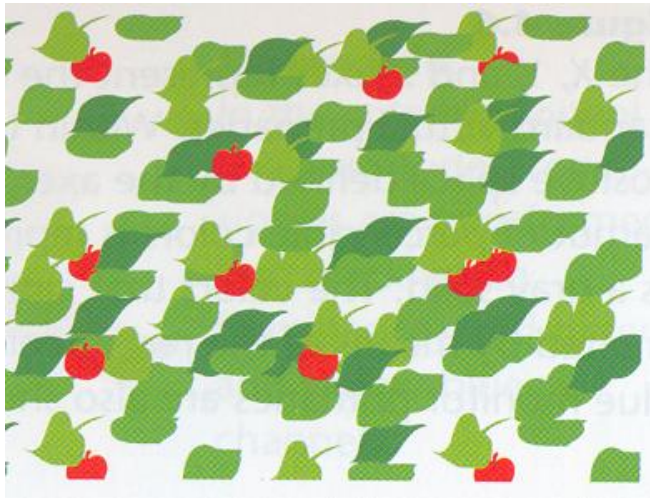


How many cherries?



(Ware, 2004)

How many cherries?



Colour may support users in many tasks!  
(yet, if not properly used may make them more difficult!)

# Using colour

Besides increasing realism, it may have the following **advantages**:

It may:

- Show the logical organization of the information displayed
- Ease the search in complex displays
- Represent approximate values
- Catch the attention
- Increase satisfaction
- Trigger emotions

...

However, **it may degrade user's performance** if not used properly

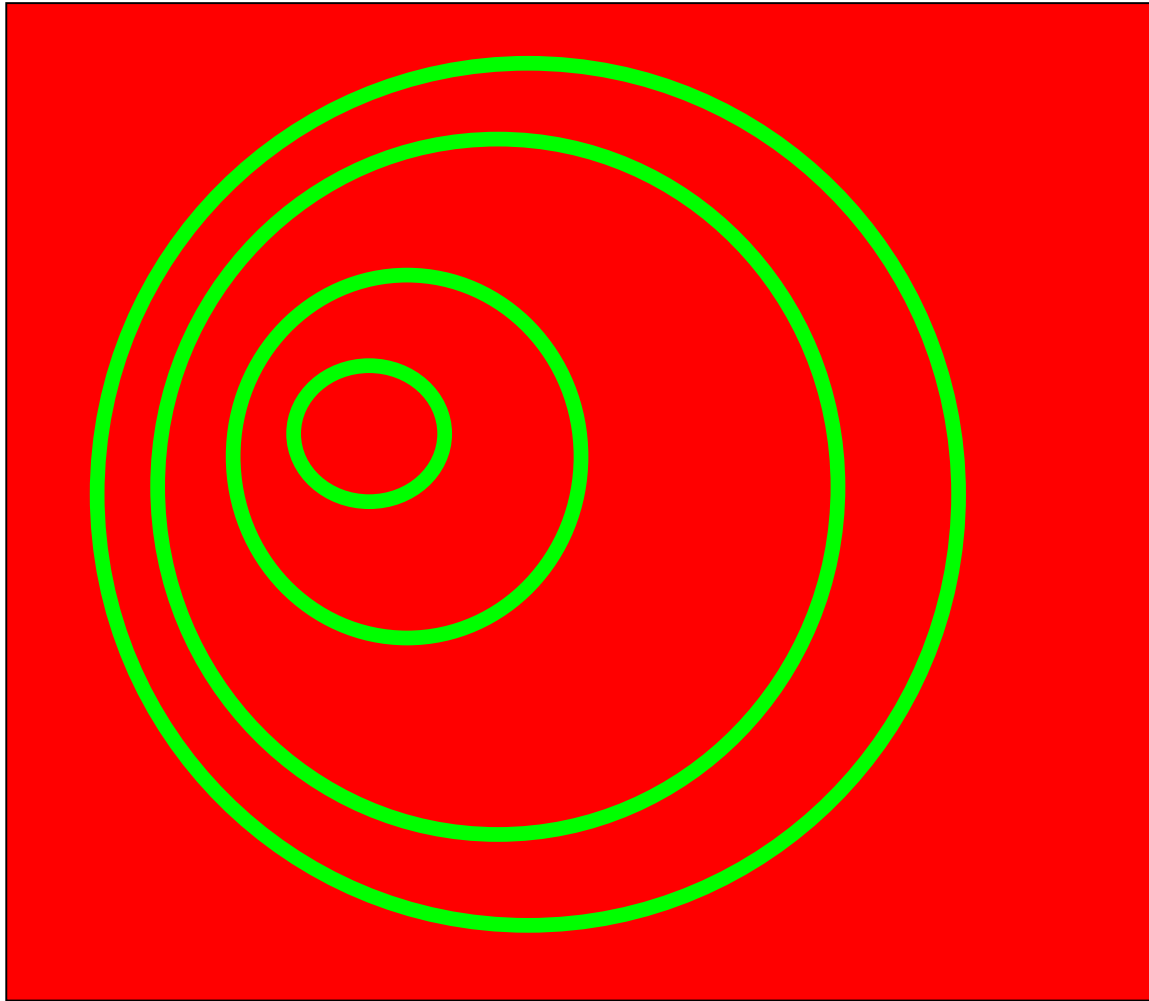
# Guidelines for using colour

- **Use colour parsimoniously**
- Use a limited number of colours
- Firstly make it work without colour
- Use colour coherently
- Avoid using simultaneously several saturated colours
- Do not convey information solely through colour
- Make colour coding support the user task
- Make the colour coding as obvious as possible
- Take into account the cultural meaning of colours ...





Take into account the cultural meaning of colours



Saturated complementary colours should not be used simultaneously

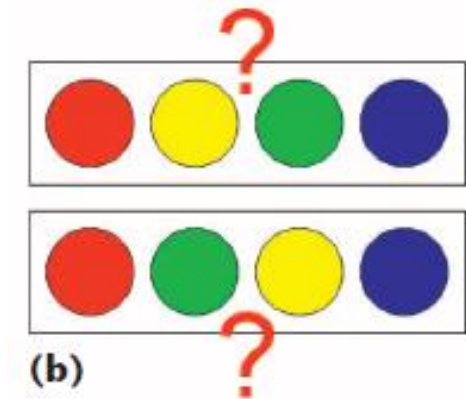
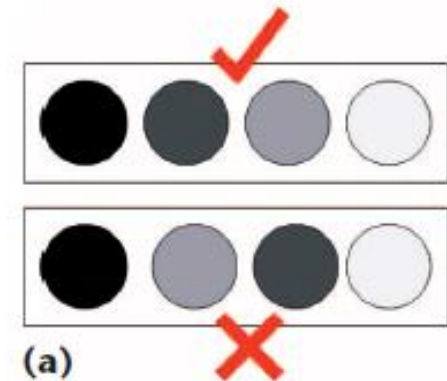


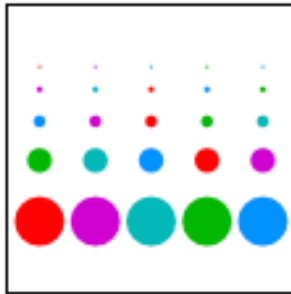
## Small spots of colour on a neutral background enhance relevant information

(Tufte, 1990) 37

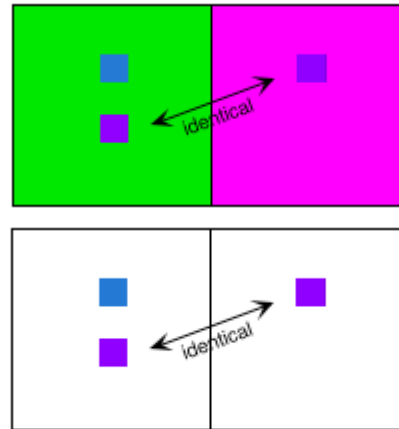
- Do not expect to perceive order from colour

(Borland, Taylor II, 2007)





Don't use colour coding on small elements



Use neutral gray surrounds where colour judgments are critical.



Both X are the same colour...

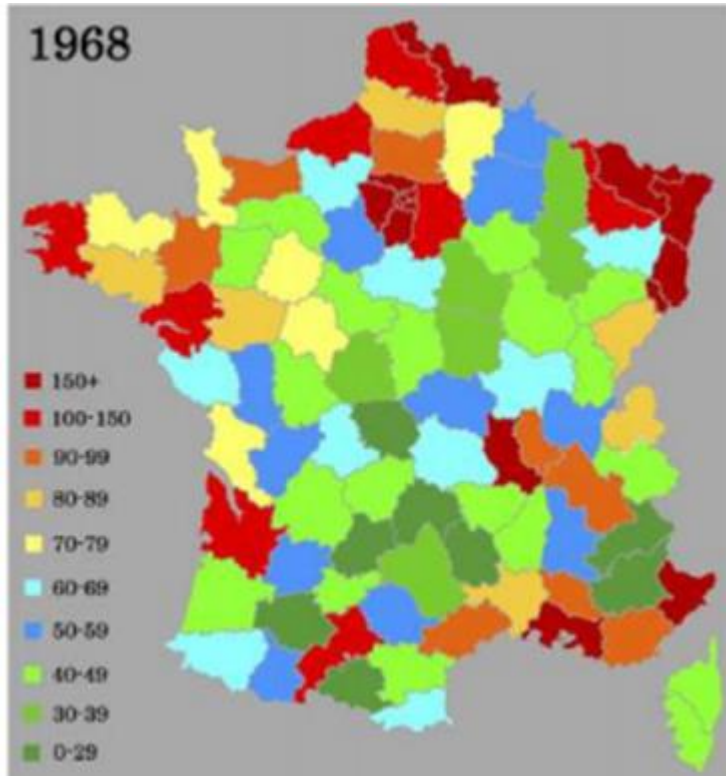
[These X's Are The Same Shade, So What Does That Say About Color? : Health News : NPR](#)



Colour may not help or even **make it more difficult!**

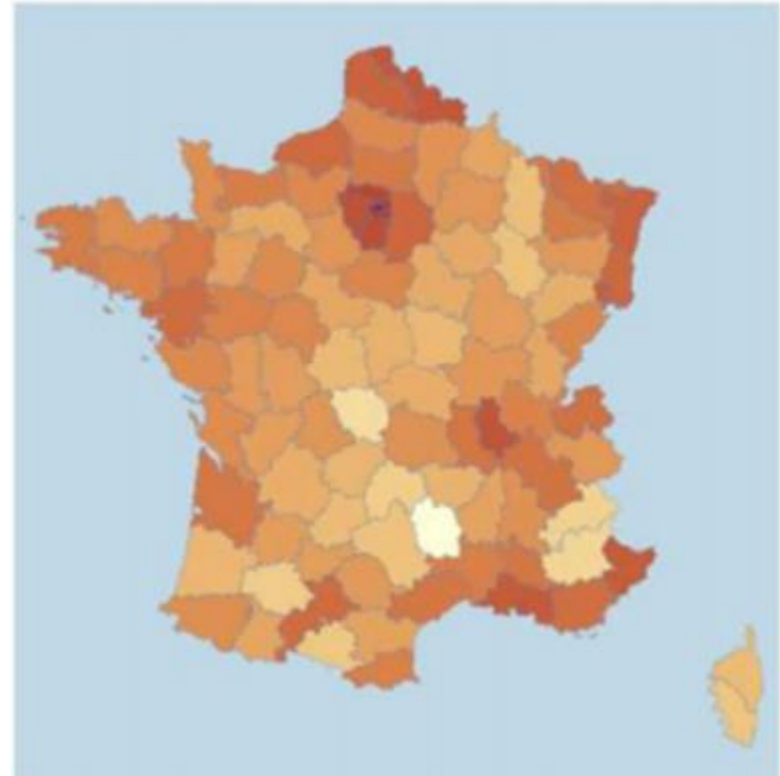
Colour scales must be carefully selected

**A**



A- no preattentive association that allows efficiently determine the values

**B**



B- a single hue and a sequential colour scheme representing values in an immediately understandable way

[The visualization anatomy – data presentation - Data Visualization: a successful design process \[Book\]](#)

# Colour scales: Qualitative colour scales to distinguish items or groups

Okabe Ito



ColorBrewer Dark2

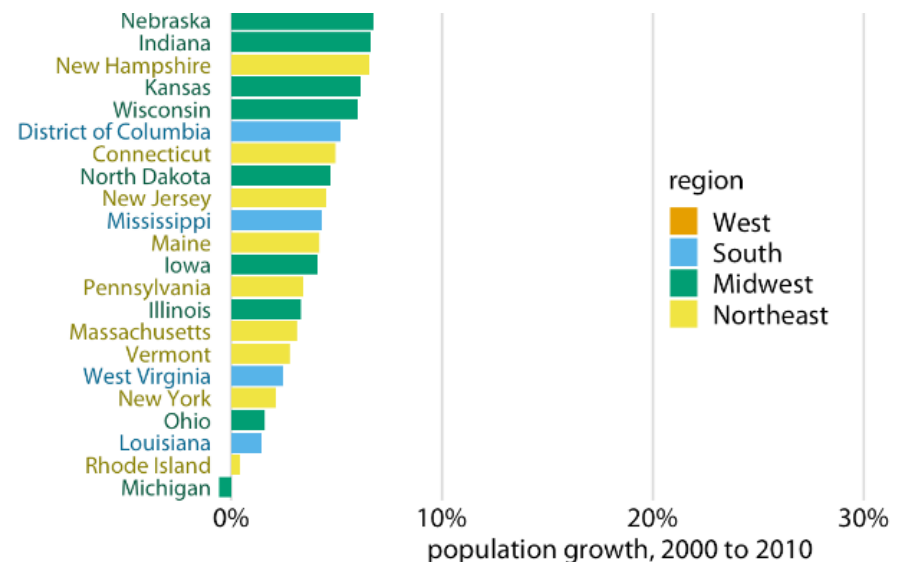


ggplot2 hue



Colours are chosen to be clearly distinct  
and not stand out relative to others

[Fundamentals of Data Visualization](#)



# Sequential colour scales to represent quantitative values

ColorBrewer Blues



Heat



CARTO Earth



Blue-Red



monochrome

diverging

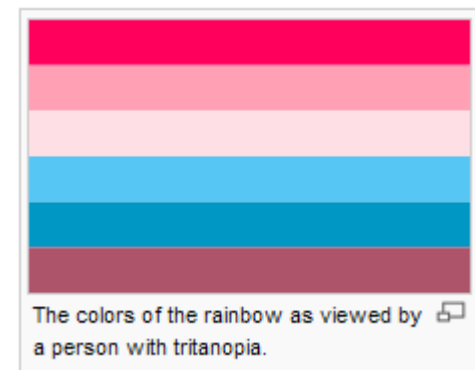


colours should indicate which values are larger or smaller, and how distant two specific values are from each other, may be monochrome, diverging ...

# Colour Vision deficiencies

- $\approx 8\%$  of men and 1% of women have some type of colour vision deficiency
- Generally it is genetic (associated to the X chromosome)
- Common deficiencies are explained by the lack of cones (colour sensor cells in the retina) sensitive to the long and medium  $\lambda$  (dicromacies):
  - Protanopia (LW – “Red” cone)
  - Deuteranopia (MW – “Green” cone) (Daltonism)
- There are three types of inherited deficiencies:
  - Monocromacy (disorder or lack of all colour sensitivity)
  - Dicromacy (disorder or lack of one type of cone)
  - Anomalous Tricromacy (disorder in cones)

# Rainbow colours as viewed by people suffering from colour vision deficiencies

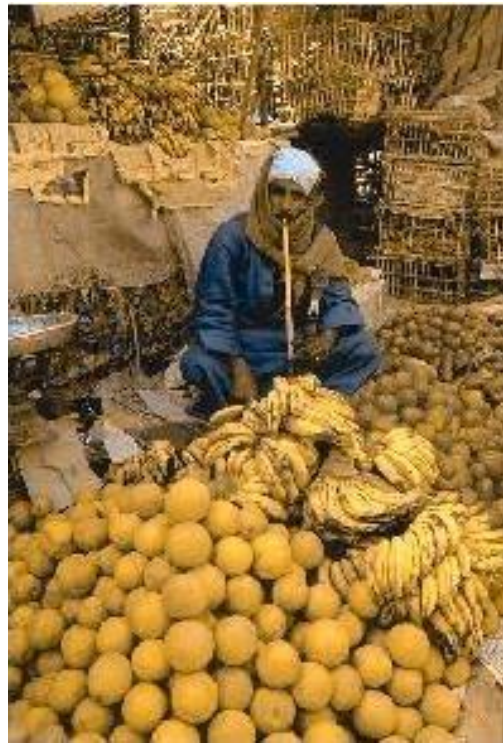


[Color blindness - Wikipedia](https://en.wikipedia.org/wiki/Color_blindness)

# Simulating colour vision deficiencies



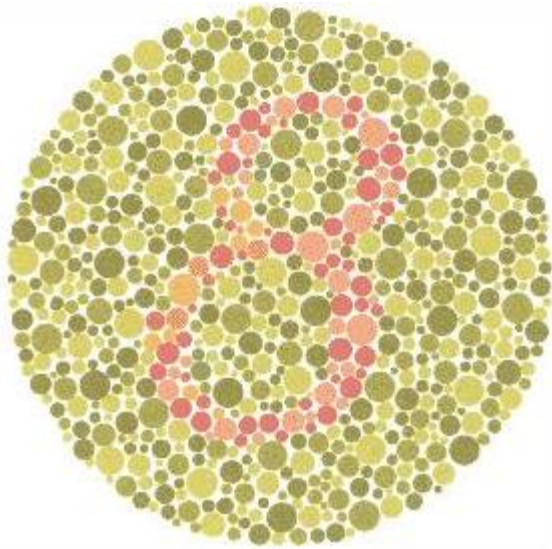
Original image as seen  
by a normal observer



As seen by a deuteranope  
(daltonic)

# Simulating colour vision deficiencies

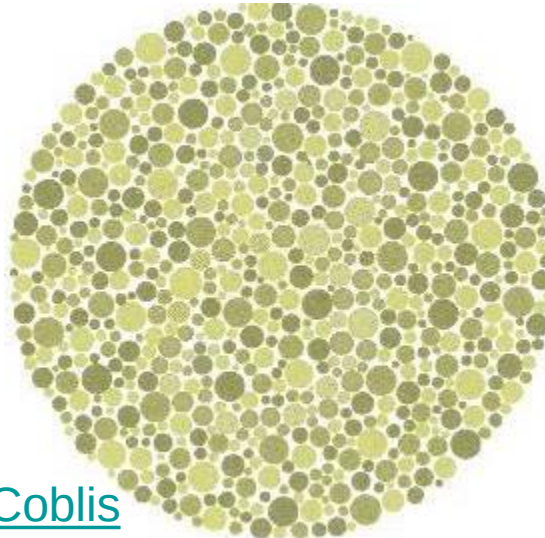
## Ishihara-2



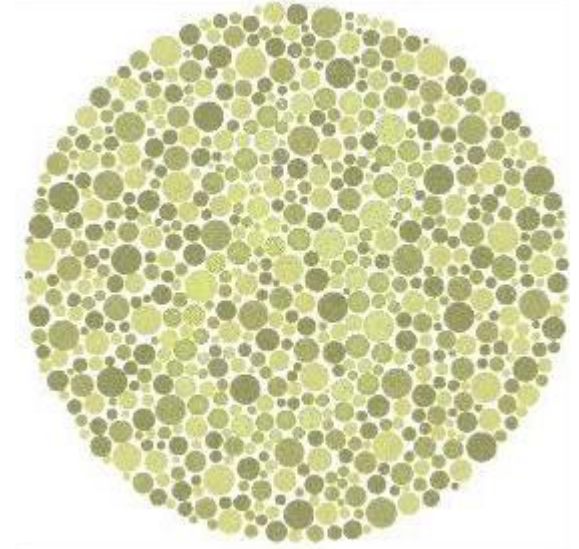
Original image as seen by a normal observer

As seen by an observer with a colour vision deficiency:

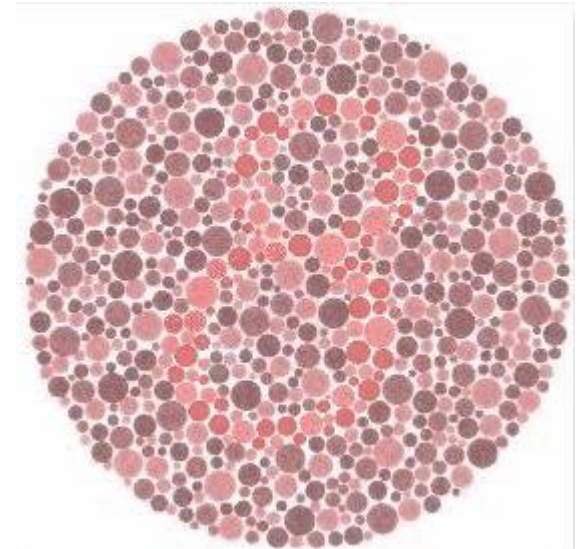
Red-Blind/Protanopia



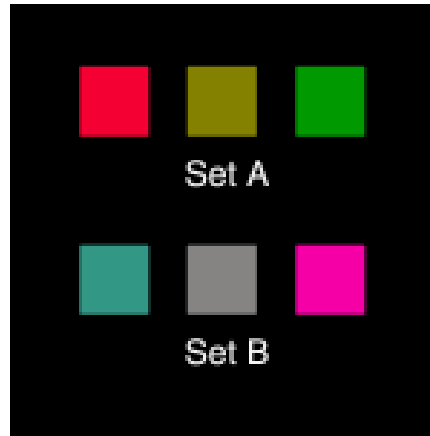
Green-Blind/Deuteranopia



Blue-Blind/Tritanopia

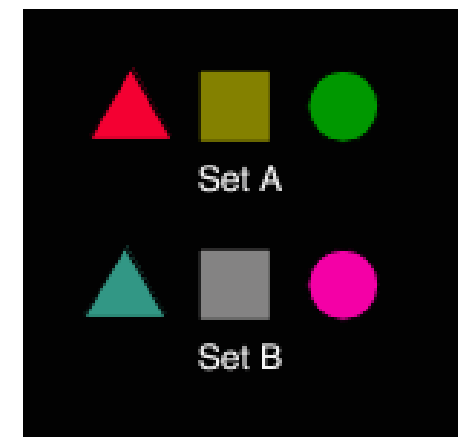


The elements within these sets look identical to deuteranopes, the most common kind of dichromat:



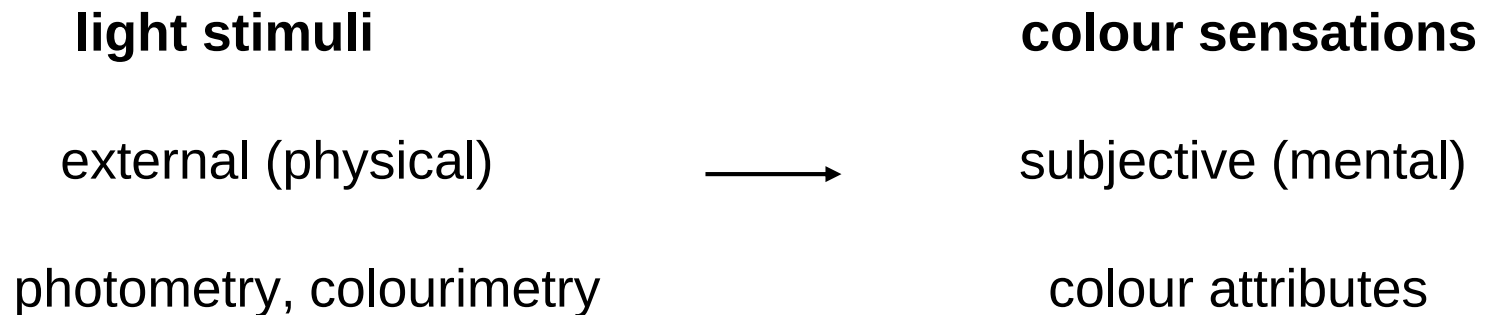
“Make it right in black and white”:  
this will work even without colour

These can be discriminated on the basis of non-color differences:



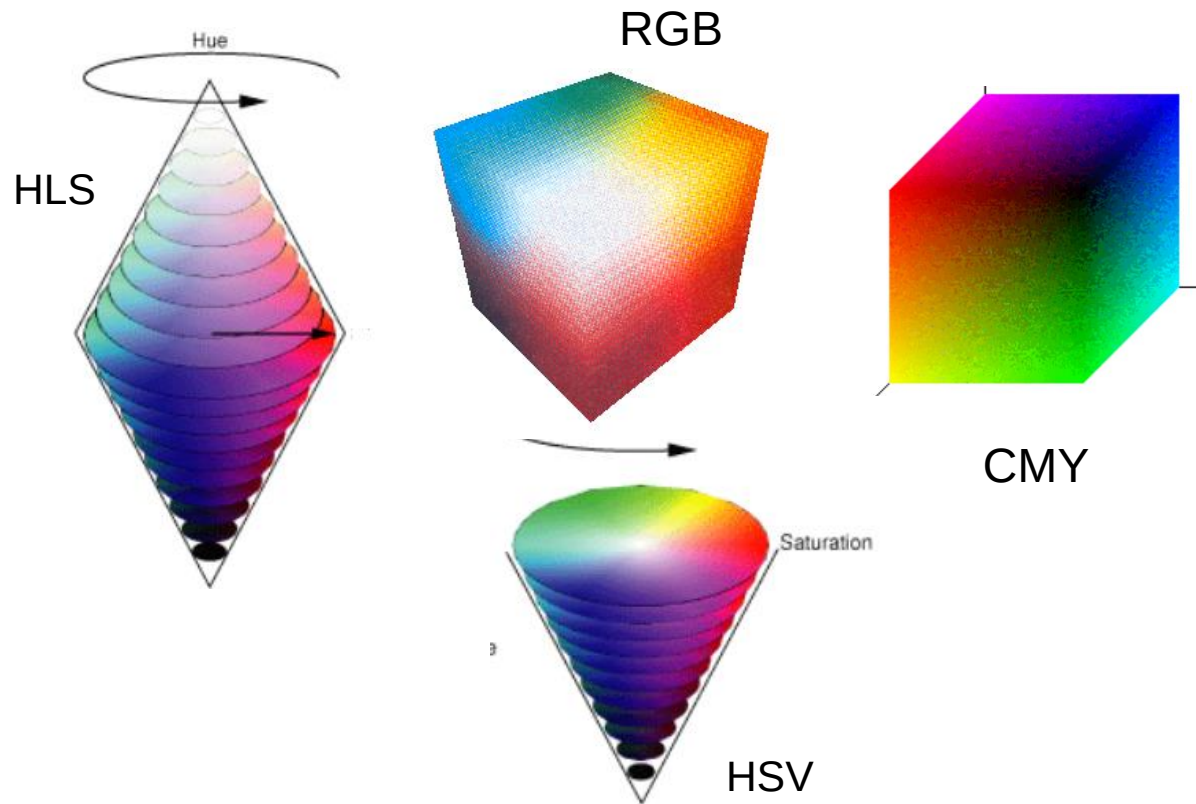
# How can we describe colour experience?

- Colour perception happens in the mind due to light properties
- Different colour descriptions are necessary for:



- **Colour models** allow describing colour objectively (very important in many situations)

# Colour Models



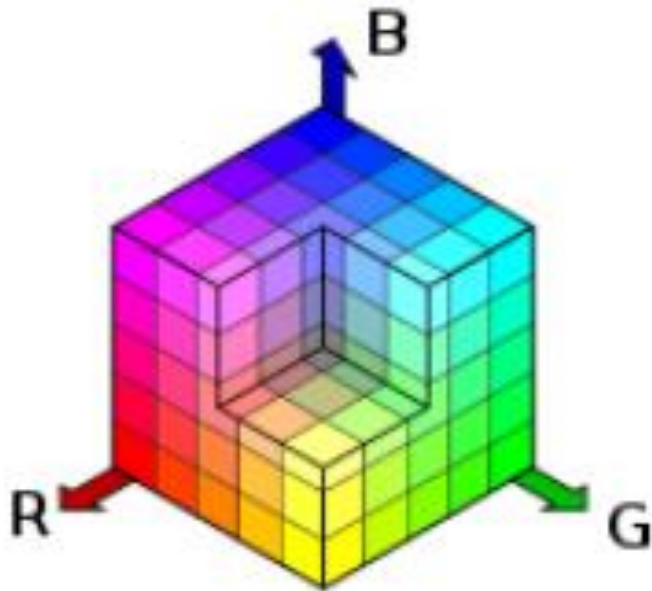
- Objects are perceived as having a colour depending on the spectrum of the reflected light (or emitted)
- But different spectra may induce similar colour sensations
- It is important to be able to describe colour objectively
- There are two types of colour production systems:
  - Additive (e.g.: monitors, TV sets, projectors) → RGB
  - Subtractive (e.g.: printers) → CMY
- **RGB** and **CMY** are **H/W oriented** colour models not adequate for users
- There are more colour models ...

The RGB colour model:



The **RGB colour model** is an additive colour model in which red, green, and blue light (the primary colours) are added to reproduce a broad array of colours.

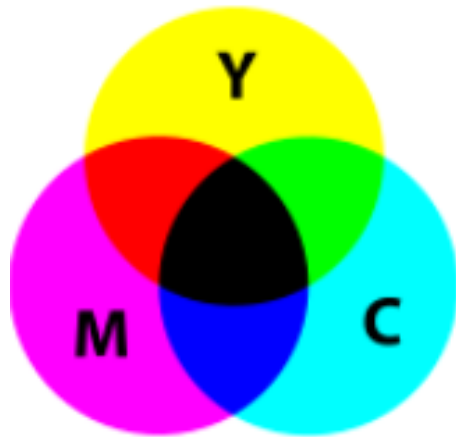
The colour space is a cube in a Cartesian coordinate system



White -> 1, 1, 1

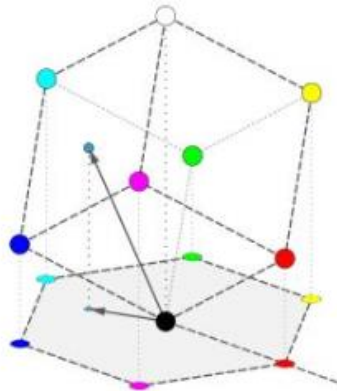
Black -> 0, 0, 0

[RGB colour model - Wikipedia](https://en.wikipedia.org/wiki/RGB_color_model)



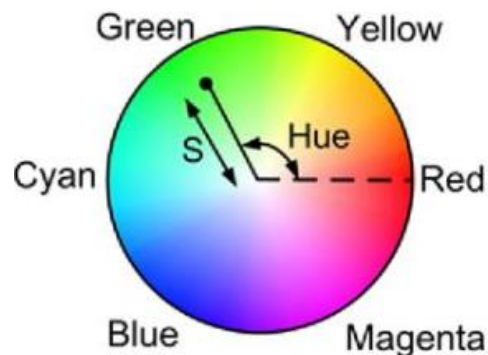
The **CMY colour model** is a subtractive colour model in which cyan, magenta, and yellow (the primary colours) are subtracted from white to reproduce a broad array of colours.

The colour space is also a cube in a Cartesian coordinate system



White -> 0, 0, 0

Black -> 1, 1, 1



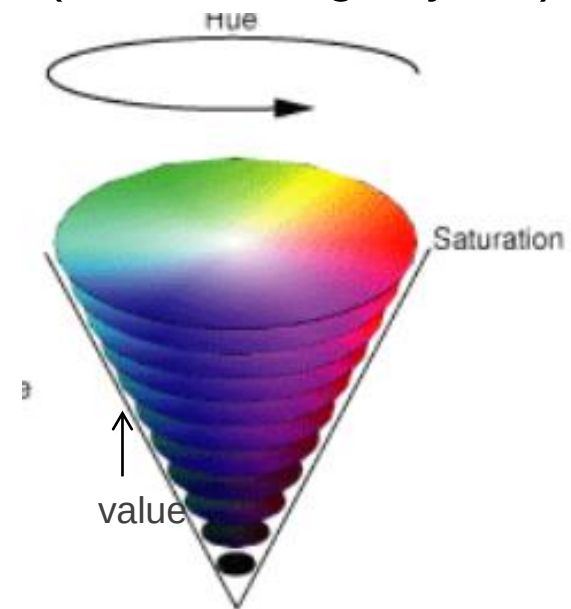
There are other models **more adequate to colour specification by the users:**

- HSV
- HLS

- Humans describe colour based on 4 **psychophysical variables** related to physical variables:
  - **Hue** – the degree to which is similar to or different from stimuli that are described as red, green, blue, and yellow
  - **Saturation** – related to the amount of achromatic light
  - **Lightness** – related to the objects reflectance (for reflecting objects)
  - **Brightness** – for light emitting objects



[Hue - Wikipedia](#)

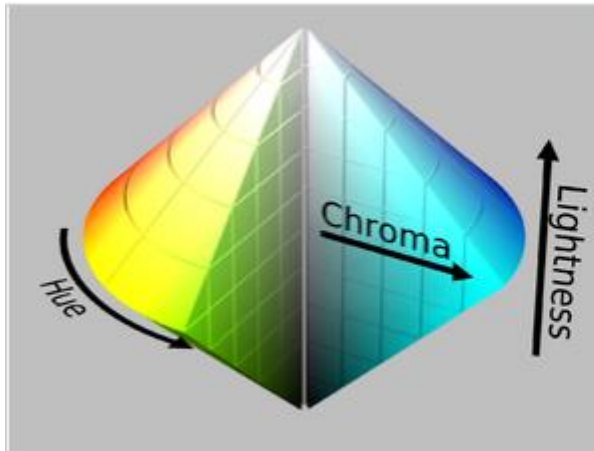


- Two colour models more adequate for the users to select colour:  
HLS and HSV

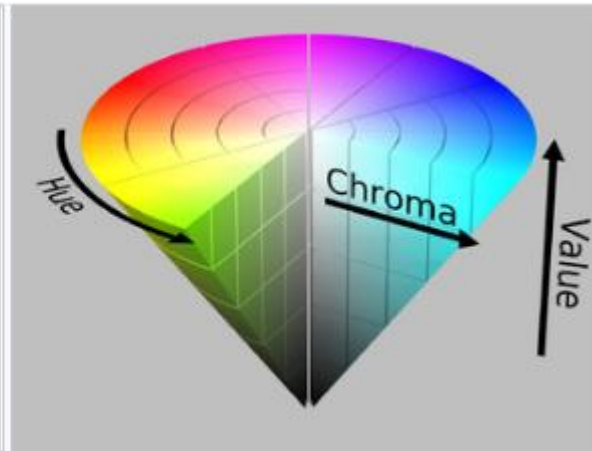
Both use cylindrical coordinates, hue and saturation

HLS: more adequate to print

HSV: more adequate to screens



HLS



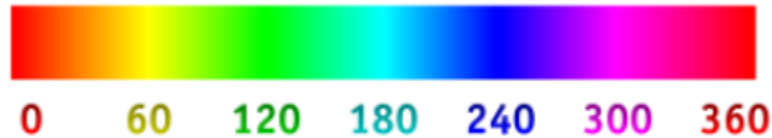
HSV

[HSL and HSV – Wikipedia](#)

[colour models and colour spaces - Programming Design Systems](#)

- HSV colour model:

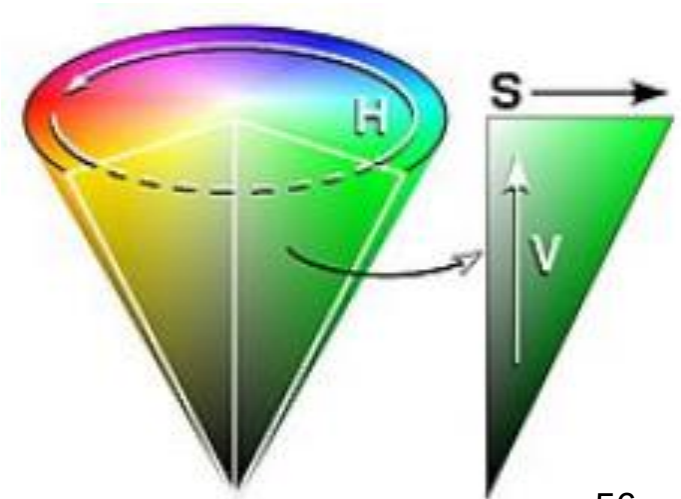
- Hue



- Saturation – related to the amount of achromatic light

- Value - controls the brightness: 0% - pure black 100% - pure white

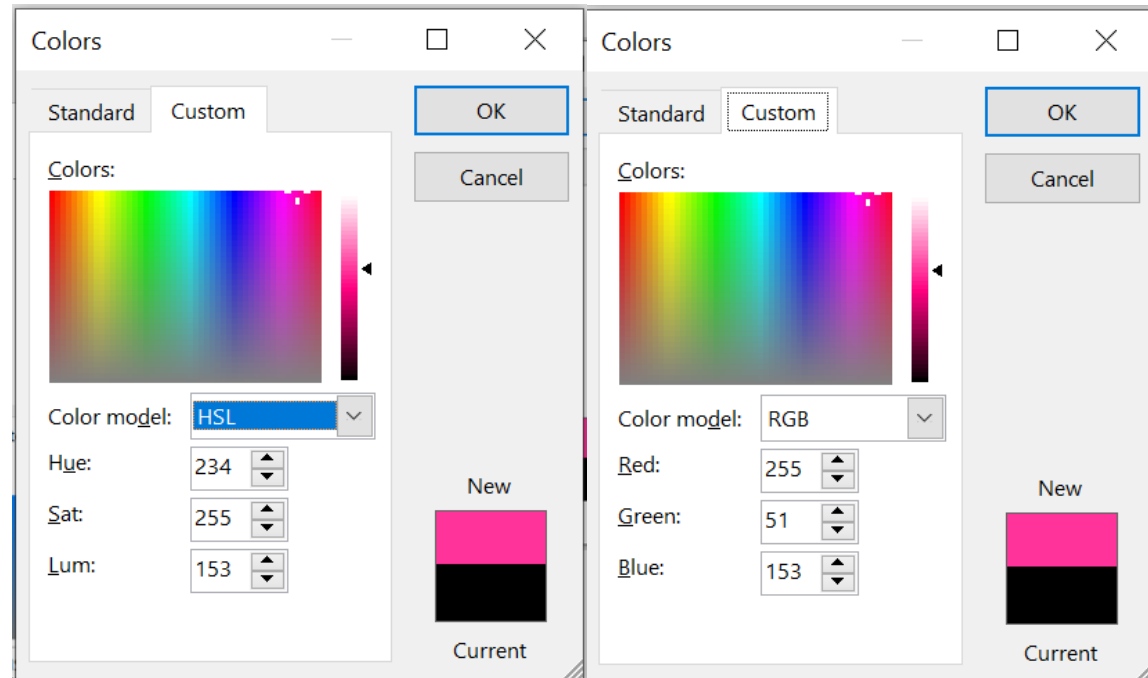
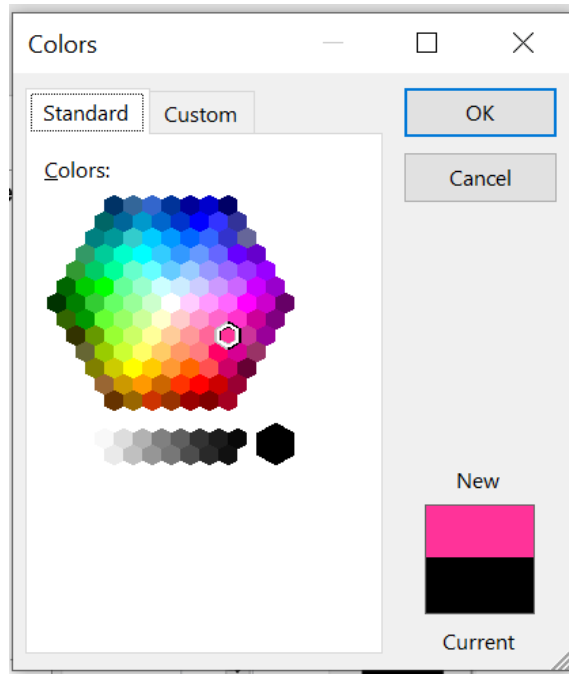
Uses cylindrical coordinates



[HSL colour model \(video\) | colour science | Khan Academy](#)

[colour models and colour spaces - Programming Design Systems](#)

- To let the user select a colour:



# Bibliography and links

J. Tidwell, A. Valencia, C. Brewer, Designing Interfaces: Patterns for Effective Interaction Design, 3<sup>rd</sup> edition, O'Reilly, 2020

[Designing Interfaces, 3rd Edition\[Book\]](#)

C. Wilke, Fundamentals of Data Visualization, O'Reilly, 2019 (Chap. 19)

[19. Common Pitfalls of colour Use - Fundamentals of Data Visualization \[Book\]](#)

[What is User Interface \(UI\) Design? | IxDF](#)

[Designing for iOS | Apple Developer Documentation](#)

[Design & Plan | Android Developers](#)

[colour Guidelines, Standards, and Requirements](#)